

CHAINAT AGROMET, Thailand

WMO: 484020

Lat: 15.1581N

Lon: 100.1914E

Elev: 55

StdP: 14.67

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	62.9	65.4	51.8	57.3	70.8	55.0	64.5	73.1	13.8	78.5	11.6	78.4	1.2	N/A	0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	17.9	99.6	78.9	97.5	79.1	95.8	79.1	82.4	92.0	81.7	91.2	81.1	90.5	3.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.1	156.8	86.6	79.5	153.3	86.0	78.9	150.4	85.5	46.3	92.2	45.6	91.6	45.0	90.7	91.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.4	10.3	7.9	59.2	102.2	3.5	2.4	56.7	103.9	54.7	105.3	52.8	106.7	50.3	108.4		
			54.8	85.7	2.9	2.3	52.7	87.3	51.0	88.7	49.3	90.0	47.2	91.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	83.4	78.4	81.9	85.1	87.7	86.8	85.4	84.2	84.0	83.7	83.4	81.7	78.8
	DBStd	3.95	3.73	3.14	3.47	2.82	3.06	2.00	1.94	1.71	1.84	2.06	3.23	3.95
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	12196	879	892	1088	1130	1140	1062	1061	1054	1011	1035	952	892
	CDD65	6723	415	472	624	680	675	612	596	589	561	570	502	427
	CDH74	76168	3648	4895	7492	9031	8639	7455	6820	6576	6163	6346	5227	3876
	CDH80	33899	1399	2154	3741	4986	4416	3369	2793	2587	2393	2479	2085	1497
Wind	WSAvg	1.7	1.1	1.4	2.1	2.2	2.1	2.3	2.0	1.8	1.1	0.9	1.3	1.5
Precipitation	PrecAvg	45.7	0.3	0.4	1.7	2.9	5.9	5.2	5.4	6.9	10.0	5.7	1.0	0.2
	PrecMax	59.3	1.4	2.1	5.4	6.5	10.2	7.8	9.6	11.4	15.0	16.3	4.1	1.8
	PrecMin	31.7	0.0	0.0	0.0	0.4	1.3	1.6	2.3	3.5	6.1	0.3	0.0	0.0
	PrecStd	6.6	0.4	0.5	1.3	1.7	2.2	1.8	1.7	2.4	3.0	1.1	0.4	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.9	95.4	100.4	103.2	102.1	97.9	96.0	94.4	94.3	93.9	94.3	93.6
		MCWB	75.5	77.4	75.5	77.6	79.8	79.7	79.8	80.0	80.9	79.2	78.0	76.7
	2%	DB	90.8	93.5	97.2	100.3	98.9	95.3	93.6	92.7	92.5	92.2	92.5	91.6
		MCWB	74.9	77.4	77.6	78.5	80.3	79.6	79.6	79.7	80.2	79.0	77.2	76.3
	5%	DB	89.3	91.9	95.4	98.0	96.6	93.5	91.8	91.4	91.1	90.9	91.1	89.8
		MCWB	74.5	77.0	77.9	78.9	80.1	79.8	79.4	79.4	79.8	78.7	76.8	74.8
	10%	DB	87.5	90.3	93.7	96.0	94.4	91.7	90.2	89.9	89.6	89.6	89.3	87.7
		MCWB	73.6	76.4	78.0	79.0	80.0	79.7	79.1	79.1	79.6	78.4	76.3	73.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.8	81.0	81.6	82.8	83.4	82.6	82.2	82.0	82.9	81.9	81.5	79.9
		MCDB	88.0	91.1	92.3	94.5	94.3	91.0	90.8	90.4	90.5	89.2	88.4	88.6
	2%	WB	78.2	80.0	80.5	81.8	82.1	81.4	80.9	81.1	81.7	81.2	80.3	78.3
		MCDB	86.5	89.8	91.8	93.1	93.0	90.5	89.9	89.6	89.3	88.3	88.2	86.8
	5%	WB	77.1	79.1	79.8	81.0	81.4	80.7	80.2	80.4	80.9	80.5	79.4	77.2
		MCDB	85.0	88.7	90.7	92.0	92.0	90.2	89.0	88.9	88.5	87.5	86.7	85.4
	10%	WB	75.9	78.2	79.1	80.3	80.7	80.1	79.6	79.7	80.1	79.8	78.3	76.0
		MCDB	83.4	87.3	89.6	91.0	91.0	89.0	88.1	87.7	87.4	86.6	85.2	84.1
Mean Daily Temperature Range	5% DB	MDBR	18.8	18.0	17.9	17.9	15.2	13.3	12.7	12.6	12.6	12.5	15.1	17.9
		MCDBR	20.1	19.2	19.9	19.8	18.1	15.7	14.9	14.7	14.5	14.2	16.6	18.8
		MCWBR	7.3	6.3	5.2	4.9	4.9	4.7	5.1	5.1	5.0	4.7	4.9	6.1
	5% WB	MCDBR	16.7	17.4	18.3	18.0	16.4	14.1	13.4	13.7	13.6	12.9	14.7	16.8
		MCWBR	6.9	6.4	5.7	5.5	5.3	5.0	5.3	5.4	5.3	5.0	5.7	6.4
Clear-Sky Solar Irradiance	taub		0.520	0.605	0.669	0.597	0.499	0.447	0.438	0.441	0.477	0.565	0.505	0.505
	taud		1.982	1.806	1.693	1.896	2.221	2.370	2.394	2.372	2.259	1.967	2.108	2.040
	Ebn at Noon		245	230	220	236	257	268	271	272	264	237	248	245
	Edh at Noon		56	69	79	64	46	39	38	40	44	58	49	51
All-Sky Solar Radiation	RadAvg		1524	1668	1745	1901	1786	1673	1575	1547	1525	1519	1537	1492
	RadStd		91	65	114	92	118	100	90	98	81	88	85	56

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (25 neighbors)	N/A	N/A	N/A	N/A	+0.34	+0.43	N/A	N/A	+167	+164

Nomenclature: See separate page